

Chapter 11: Finding the Unknown

Worksheet 11: Finding the Unknown

Learning Objective: Understand the structure of an equation, solve linear equations in one variable using the balancing-and-transposition method, and form equations from word problems.

Worked Example:

Solve for x : $3x + 5 = 20$.

Solution: Subtract 5 from both sides: $3x = 15$. Divide both sides by 3: $x = 5$.

Questions (1–20)

1. Solve: $x + 7 = 12$.

2. Solve: $5x = 35$.

3. Solve: $x - 9 = 14$.

4. Solve: $\frac{x}{4} = 6$.

5. Solve: $2x + 3 = 11$.

6. Solve: $3x - 4 = 17$.

7. Solve: $4x + 7 = 3x + 12$.

8. Solve: $2(x + 3) = 16$.

9. A number increased by 8 gives 25. Form an equation and find the number.

10. Solve: $5x - 3 = 2x + 12$.

11. Solve: $3(x - 2) + 4 = 19$.

12. The sum of a number and twice itself is 36. Form an equation and find the number.

13. A father is 3 times as old as his son. If the sum of their ages is 48 years, form an equation (son's age = x) and find both ages.

14. Solve: $\frac{2x+5}{3} = 7$.

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15. Three consecutive numbers add up to 72. Form an equation (first number = x) and find the numbers.
16. A number is such that twice the number, minus 5, equals 3 more than the number itself. Form an equation and solve for the number.
17. A rectangle's length is 5 cm more than its width. If the perimeter is 50 cm, form an equation (width = x) and find the length and width.
18. Two numbers differ by 12, and their sum is 48. Form an equation and find both numbers.
19. Your age 5 years from now will be 20. Form an equation to find your current age, and solve it.
20. A shopkeeper sells pens at Rs. x each. Selling 6 pens gives Rs. 90. Form an equation and find the price of one pen.

Reflection Question: Why is it important to do the same operation to both sides of an equation when solving it? Explain in your own words.

Real-Life Application Question: Think of a real situation involving an unknown quantity (your savings, a friend's age, a distance). Write it as an equation using a letter, and try to solve it.